

Year 11 Applications
Term Three Holiday Work
Univariate Data, Normal Distribution & Trigonometry

Univariate Data – Calculator Free

1. Of a set of numbers, 6 are given below:

4 4 5 5 5 6

What must the seventh number be so that for the seven values:

- (a) The mean is equal to the median.
- (b) The mean is equal to the mode.

2. The table below shows the sales of men's shoes at a local market.

Shoe Size	9	10	11	12	13	14
Frequency	3	5	7	1	2	2

- (a) Calculate the mean, median and modal shoe size.
- (b) Which measure of central tendency is of most value to the owner of the shoe store?

Univariate Data – Calculator Assumed

4. During a term, a student has an average of 56% after the first four tests and his average for the next six tests is 44%. If the student does not have an overall mean of at least 50%, he must repeat the unit. Will the student have to repeat? Provide evidence to support your answer.

5. George owns a small independent supermarket. Last year, he paid himself \$325 000 and his employees were paid as follows:

Position	Number of Employees	Annual Salary
Owner	1	\$325 000
Manager	1	\$85 000
Assistant Managers	3	\$65 000
Accountant	1	\$58 000
Marketing Manager	1	\$52 000
Fresh Produce Supervisors	3	\$38 000
Butcher	2	\$34 500
Retail Assistants	10	\$31 000
Security Guard	2	\$29 500

- (a) Determine the mean, mode and median salary of the workers in this company.
- (b) One of the employees argues that the workers receive a poor average wage compared to other companies of a similar size. Which “average” figure would he use to support his argument?
- (c) The owner of the company argues that the workers have an excellent “average” pay. Which figure would he quote to support his argument?

Collecting, Representing & Interpreting Data – Calculator Free

6. The following information was collected from patrons working out at a local gymnasium. Classify the data as categorical ordinal, categorical nominal, numerical discrete or numerical continuous.

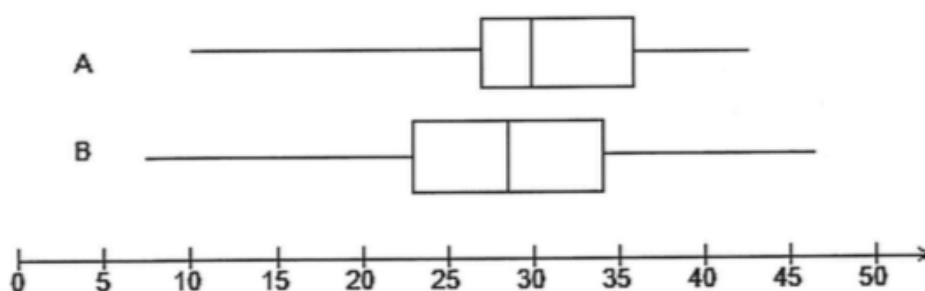
- (a) Gender
- (b) Brand of shoe worn
- (c) Height
- (d) Occupation
- (e) Number of visits to the gym in the last 7 days
- (f) Rating of the facilities as excellent, good or poor.

7. Consider the set of scores shown below.

2, 3, 3, 4, 4, 5, 8, 12

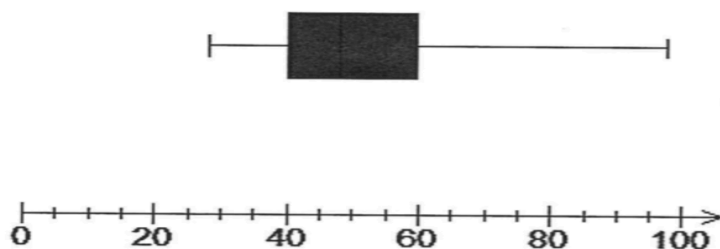
- (a) Construct a box-and-whisker plot for the data given.
- (b) Describe the shape of the distribution
- (c) Determine whether an outlier is present for the data above.
- (d) What effect will the removal of outliers have on the shape of the box-and-whisker plot.

8. Two classes, A and B, sat an identical test and the results are displayed below. Which class performed better in the test?



Collecting, Representing & Interpreting Data – Calculator Assumed

9. The boxplot below show the marks for group of 180 people in a company safety test.



(a) If an employee scored 60%, how can he use the box plot to determine how many of his colleagues scored better than him?

(b) An employee compares the whiskers of the box plot and remarks “Fewer employees scored between 25% and 40% compared employees scoring 60% to 100%. Comment on the validity of this statement.

10. In a cricket team, one batsman must lose his spot when the captain returns from injury. The last six scores of two batsman are taken into consideration.

Batsman A: 3, 5, 10, 7, 169, 10

Batsman B: 27, 36, 44, 21, 29, 38

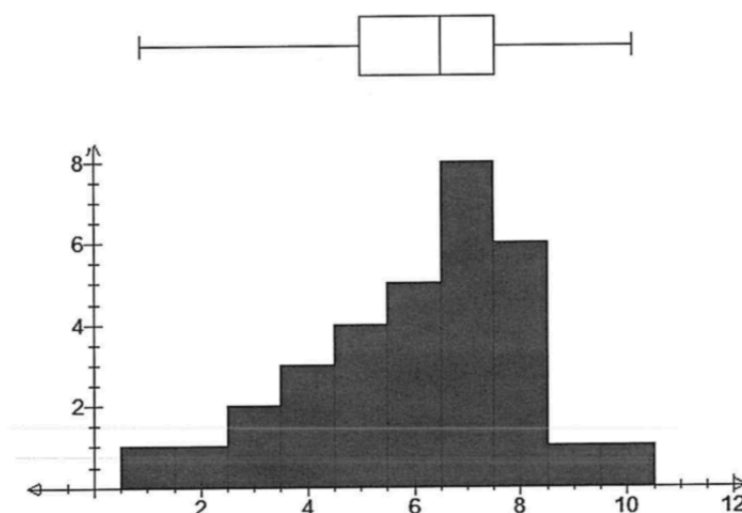
(a) Describe the nature of the data that has been collected.

(b) Calculate the respective means of the two batsman.

(c) Construct a box and whisker plot for each of the two batsman.

(d) Compare the distribution of the scores and decide who should lose their place in the team. Give reasons for your answer.

11. Consider the frequency histogram and its associated boxplot shown to the right. It shows the tooth length (to the nearest millimetre) in a particular species of fish.



- (a) Calculate the mean of the scores shown above.
- (b) Explain why your answer to (a) is only an estimate of the mean.
- (c) Describe the distribution (location, spread, shape) of data provided.

Standard Deviation and Making Judgements Between Data Sets – Calculator Free

12. Consider the three stem-and-leaf plots shown below.

A		B		C	
Stem	Leaf	Stem	Leaf	Stem	Leaf
3		3	0 0	3	5
4	5	4		4	
5	0	5	0	5	0
6	0 0	6	5	6	
7		7		7	0 0

Which of the stem-and-leaf plots would you expect to:

- (a) Have the smallest standard deviation?
- (b) Have the lowest mean?

13. The following scores have a standard deviation of 1.51 (2 dp.).

2, 3, 3, 4, 4, 4, 5, 7

- (a) What effect, if any, will the addition of a score of 11 have upon:
 - (i) The mean
 - (ii) The standard deviation.

Standard Deviation and Making Judgements Between Data Sets – Calculator Assumed

14. Quinn's performances in his Italian & Japanese test are shown below:

Italian: 58%, 55%, 49%, 51%, 63%, 65%, 71%, 46%

Japanese: 56%, 84%, 87%, 42%, 37%, 69%, 51%, 72%

(a) Determine which subject Quinn achieved the highest mean in and state this mean.

(b) In which subject was Quinn more consistent?

15. A 200 gram packet of "N & N"s contains approximately 210 chocolates candies. A small factory has two machines that fill the bags and seal them for distribution to retailers. 25 packets from each machine were chosen at random and their contents were recorded in the stem and leaf plot below.

Machine 1			Machine 2	
	1	20 _L	4	
8 8 8 8 8 8 8 8 8		20 _H	6 8 8 8 8 8 9 9	
3 3 2 2 1 1 0 0 0 0		21 _L	0 0 0 0 1 1 1 2 2 2 2 3	
	5 3	21 _H	6 6 7	
	7	22 _L	0	
		22 _H		

(a) Calculate the mean and standard deviation of the number of chocolates candies inserted by each of the machines.

(b) Based on the 25 samples:

(i) Which of the two machines is more cost effective?

(ii) Which of the two machines is more consistent?

Normal Distribution – Calculator Free

16. A manufacturing company approximates the length of its matchsticks using a normal distribution model. The matchsticks have a mean of 4.5cm and a standard deviation of 1 mm.

For a randomly selected matchstick, use the 68%, 95%, 99.7% rule to determine the probability that it measures:

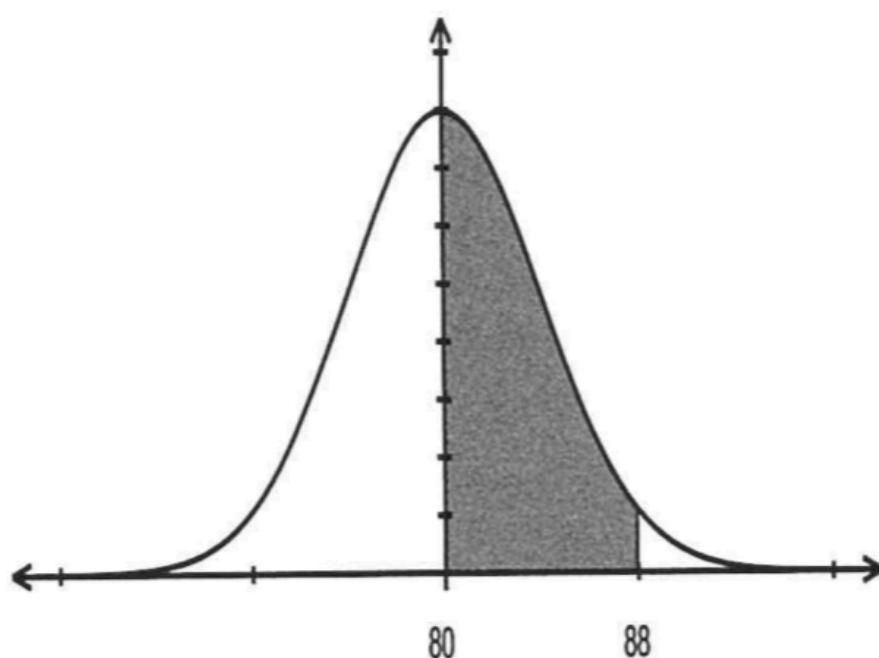
- (a) Between 4.4cm and 4.6 cm
- (b) Between 4.3cm and 4.7cm
- (c) Greater than 4.6 cm.

17. The standby time on a new model of smartphone is normally distributed. The mean standby time is 500 hours with a standard deviation of 18 hours. Use the 68%, 95%, 99.7% rule to determine what percentage of this particular model of smartphone has a standby time longer than 536 hours.

18. If $X \sim N(12.5, 3^2)$, find, using the 68%, 95%, 99.7% rule:

- (a) $P(X < 15.5)$
- (b) $P(6.5 < X < 15.5)$
- (c) $P(9.5 < X < 15.5)$

19. A set of scores is normally distributed with a mean of 80. If the shaded area on the diagram below is approximately 0.475, what is the standard deviation for this set of scores?



Normal Distribution – Calculator Assumed

20. The contents of a particular brand of tinned soup is normally distributed with a mean of 250 mL and a standard deviation of 8 mL. Determine the probability that a randomly selected container tin contains between 250 & 260 mL of soup.

21. A coffee machine claims to dispense 150 mL of the selected beverage. The volumes of the drinks dispensed to be normally distributed with a mean of 149.5 mL and a standard deviation 1.4 mL.

- (a) What proportion of drinks will be between 149 and 151 mL?
- (b) What is the least that can be expected in the largest 5% of beverages?

22. Julia recently completed her end of year exams. Julia finds out from her languages teachers that the Italian exam had a mean of 62% and a standard deviation of 12 and the French exam had a mean of 48% and a standard deviation of 8. In which subject did Julia attain the “better” result? Justify your answer.

23. A new experimental drug undergoes its first round of clinical trials. The time taken for patients to feel its effects are found to be normally distributed with a mean of 20 minutes and standard deviation of 4 minutes.

- (a) What percentage of patients take longer than 15 minutes to feel the effects of the drug?
- (b) What percentage of patients takes between 15 and 25 minutes to feel the effects of the drug?

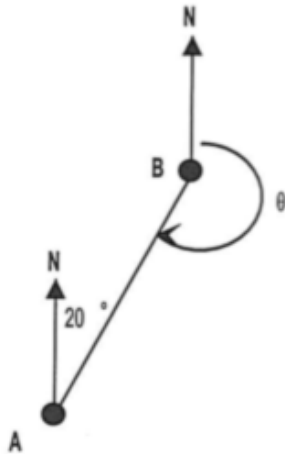
The clinical trial found that some patients were “immune” to the experimental drug and experienced no effect after taking the drug. It was determined that if a patient did not react to the drug after 35 minutes, then they are deemed to be “immune” to its effects.

- (c) According to the data, how many patients in 100 000 will be immune to the effects of the drug?

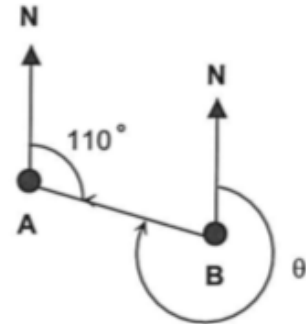
Trigonometry – Calculator Free

24. Calculate the bearing of A from B in the following two scenarios.

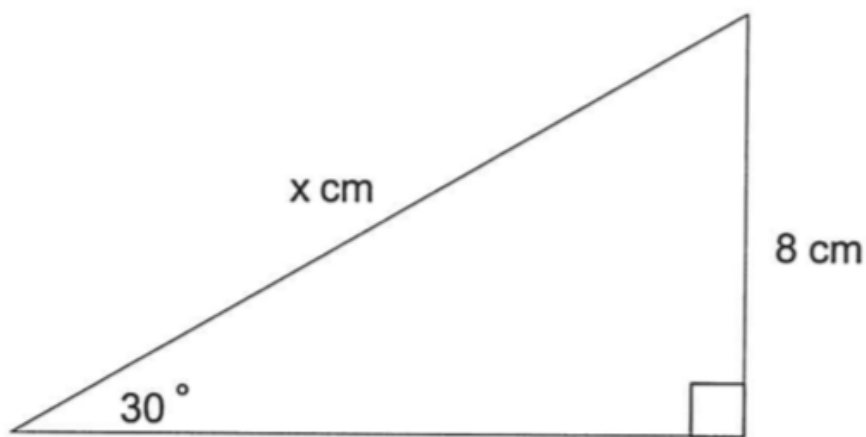
a)



b)



25. If $\sin(30^\circ) = 0.5$, calculate the missing side of the triangle below.

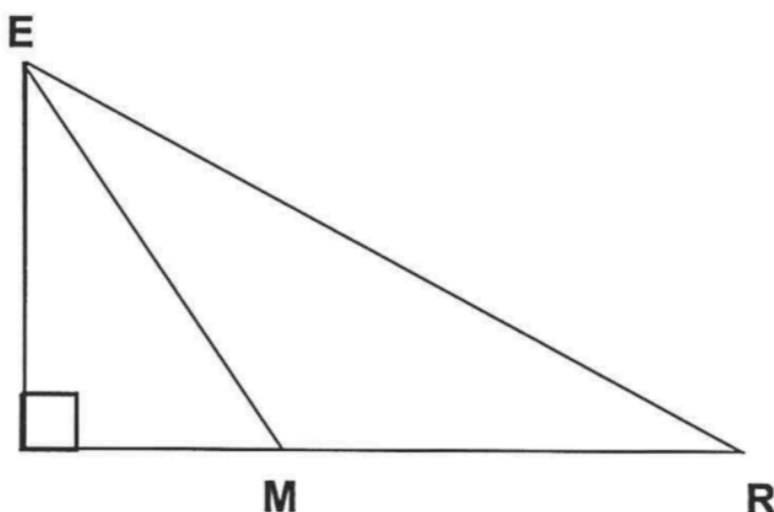


Trigonometry – Calculator Assumed

26. A 53 metre wire attached to the top of a pole attaches to a fixed point on the ground where the angle of elevation is 24° . Determine the height of the pole.

27. John leaves camp and hikes for 3.2km on a bearing of 167° . How far south of the camp has he travelled?

28. **Very Difficult:** From the top of a cliff, a perched eagle (E) spots a mouse (M) and a rat (R). The angle of elevation from the mouse to the eagle is 55° and the angle of elevation from the rat to the eagle is 27° . The mouse and the rat are 425 metres apart.



(a) What is the direct distance that the eagle must fly to catch the stationary mouse?

(b) What is the direct distance that the eagle must fly to catch the stationary rat?

(c) Determine high above the ground is the eagle perched?

29. From a 9.5 metre observation tower, a lifesaver sees a shark. He measures the angle of depression of his line of sight a 2° . Two minutes later, the angle of depression is 4° and his line of sight is in the same direction. How far has the shark swum in two minutes? Give your answer to the nearest metre.